Vast Challenge 1 Progress 1

Contributions

Robert Czarnota: Developed program to find vehicle paths/ created graphs to find patterns in non-camper traffic. (Slides 7-10)

Matthew Woods: Aggregated park service vehicle sensor data to establish a baseline ranger patrol pattern of life. (Slide 11)

Chisomaga Acholonu: Identification of Suspicious Heavy Truck Activity at Ranger-Stop Locations (Slide 13)

Tiffany Liu: Cross referenced restricted gates and rangers stops with vehicle activity to spot unusual activity (Slide 12)

Rionna Mcleary: Discovered an abnormality from a car bypassing through a gate without being detected by the sensor (Slide 14)

Introduction

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Background

2.
Explaining the Data

3.
Analysis Results

4.
Next Steps

Background

Boonsong Lekagul Nature Preserve

- Open 24 hours
- Hiking
- Camping
- Cycling
- Conservation Study

Data

- .csv file containing data recorded from sensors around the Boonsong Lekagul Nature Preserve.
- Map containing the locations of roadways and sensors throughout the Preserve.

Ground Truth

There is a truck that visits a Ranger Station in a fairly remote part of the Preserve periodically throughout the dataset timeline that is illicitly dumping factory waste. It is registering in the data as a non-Ranger vehicle, visiting a Ranger-Stop. The dumping has a slight effect on the visits to the campground nearby.

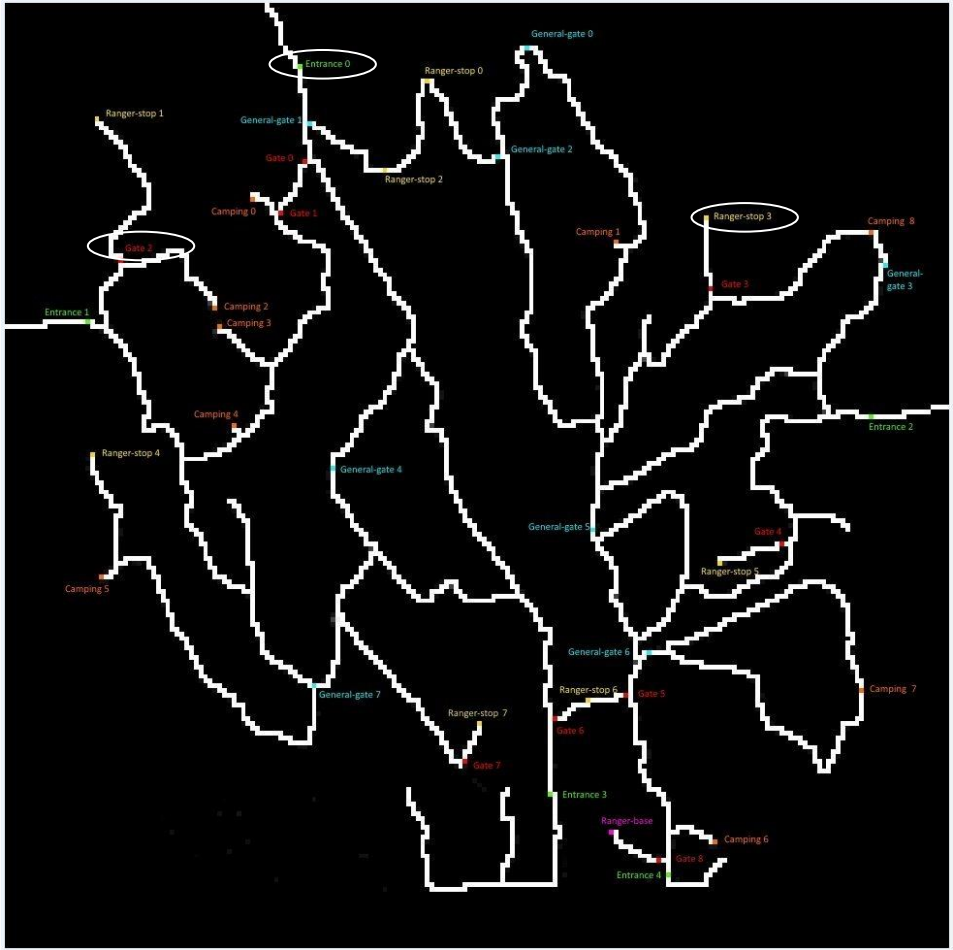
Lekagul Sensor Data

- The sensor data given holds 4 pieces of information
  - **Timestamp**
    - The exact date and time a vehicle passes through a gate.
  - **Car-id**
    - Unique identifier given to a vehicle when it enters the Lekagul Preserve.
    - The tag must be returned when a vehicle exits the preserve.
    - RF tag that is scanned when a vehicle passes through any gate in the reserve.
  - **Car-type**
    - Identifies the type of car.
    - Vehicles are designated as 1-6 and an additional designation of “P” added to ranger vehicles.
  - **Gate-name**
    - Which gate the vehicle passes through.

Timestamp	car-id	car-type	gate-name
5/1/2015 0:43	20154301124328-262	4	entrance3

Lekagul Map

- We were also given a map of the Lekagul Preserve.
- Each gate is labeled on the map.
- You can see the paths leading to each gate as well as the entrances/exits to the preserve.
- Map shows an area of 12 miles x 12 miles.



Pattern of Life: Average Time Spent in the Reserve by Non-Campers

- **Making new sessions**
  - Create a csv that groups vehicle ID by session.
  - When a vehicle enters the preserve through any entrance gate, that is the start of the session and when the same vehicle exits through an entrance gate, that is the end of a session.
- **Average time spent in the reserve:**
  - Create a calculated field in Tableau that takes the DATEDIFF between start time and end time in hours.
- **Campers v. non-campers**
  - Create a filter in Tableau that checks if the word “camp” is in Routes

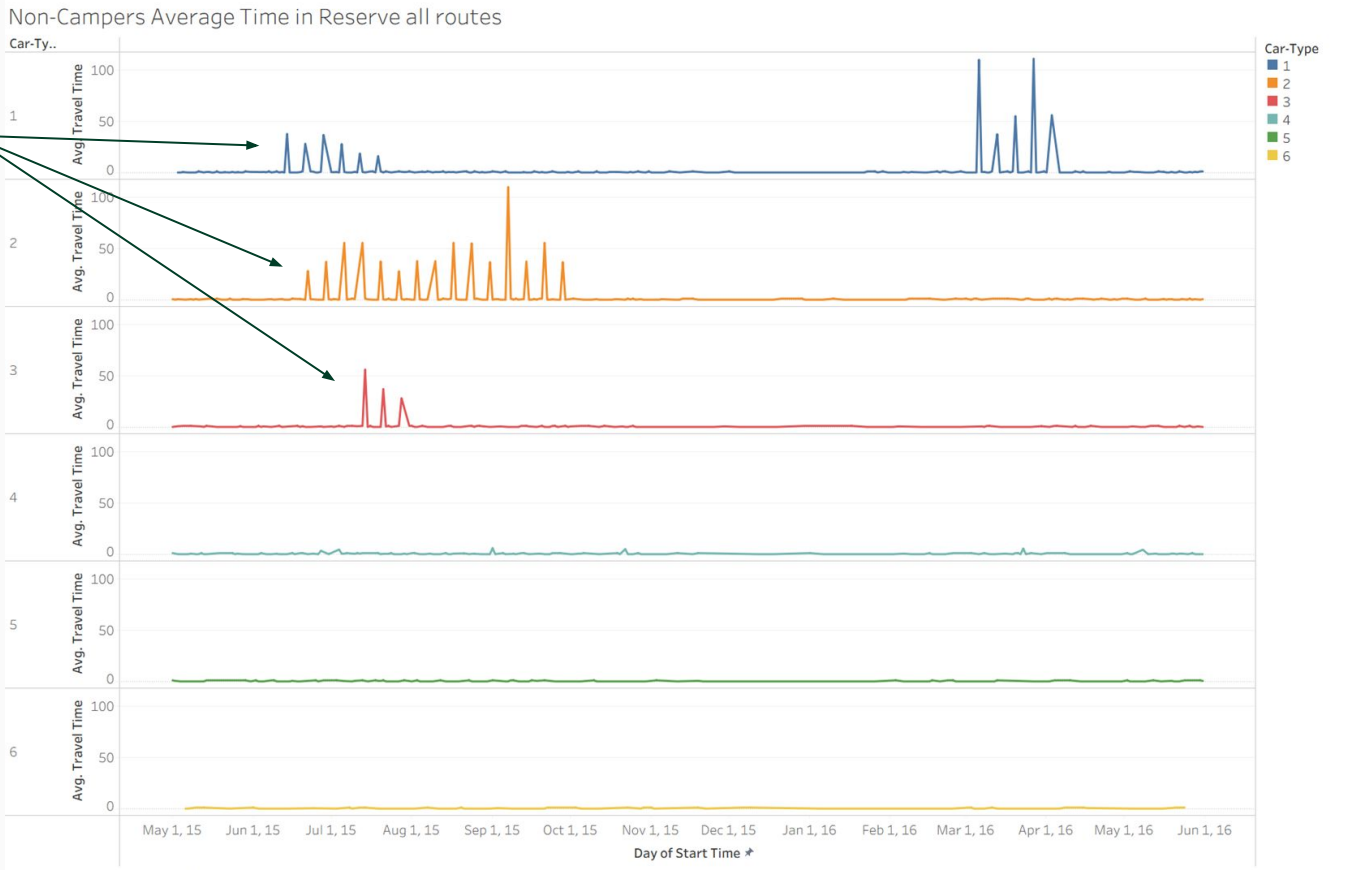
Car-id	car-type	start time	end time	route traveled
20154901054932-304	3	5/1/2015 5:49	5/1/2015 6:26	entrance1->camping2->camping2->general-gate7->entrance3

Pattern of Life: Average Time Spent in the Reserve by Non-Campers

From June 14, 2015 to there was a drastic increase in avg. time spent in the park for 3 different care types.

The pattern was consistent for all car types but the duration of that pattern varied.

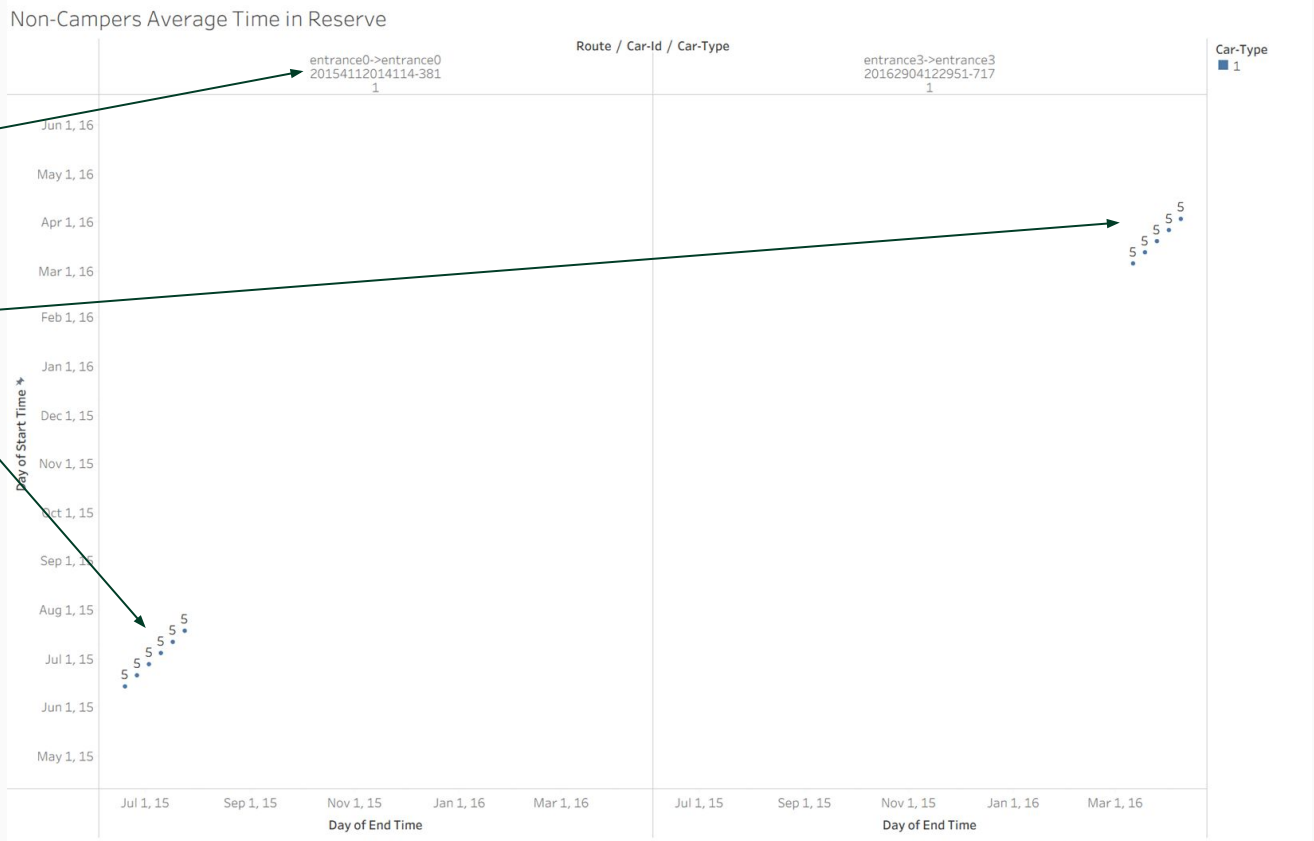
The bottom squares represent the baseline of traffic through the park. The average time spent was just under and hour.



The trend of average of Travel Time for Start Time Day broken down by Car-Type. Color shows details about Car-Type. The data is filtered on Camping and Route. The Camping filter keeps False. The Route filter keeps 9 of 281 members. The view is filtered on Car-Type, which keeps multiple members.

Pattern of Life: Average Time Spent in the Reserve by Non-Campers

- A single vehicle ID was used throughout the entire period.
- The car entered at around 11pm on Sunday and left around the 11am Friday.
- Similar pattern for all vehicles.
- The vehicles only passed through the entrance gates during their stay.



End Time Day vs. Start Time Day broken down by Route, Car-Id and Car-Type. Color shows details about Car-Type. The marks are labeled by sum of Travel Time. Details are shown for Start Time. The data is filtered on Camping, which keeps False. The view is filtered on Car-Type, Route, Inclusions (Car-Type,DAY(Start Time)) and Inclusions (Car-Type,DAY(End Time)). The Car-Type filter keeps 6 of 6 members. The Route filter excludes entrance4->entrance4. The Inclusions (Car-Type,DAY(Start Time)) filter keeps 11 members. The Inclusions (Car-Type,DAY(End Time)) filter keeps 11 members.

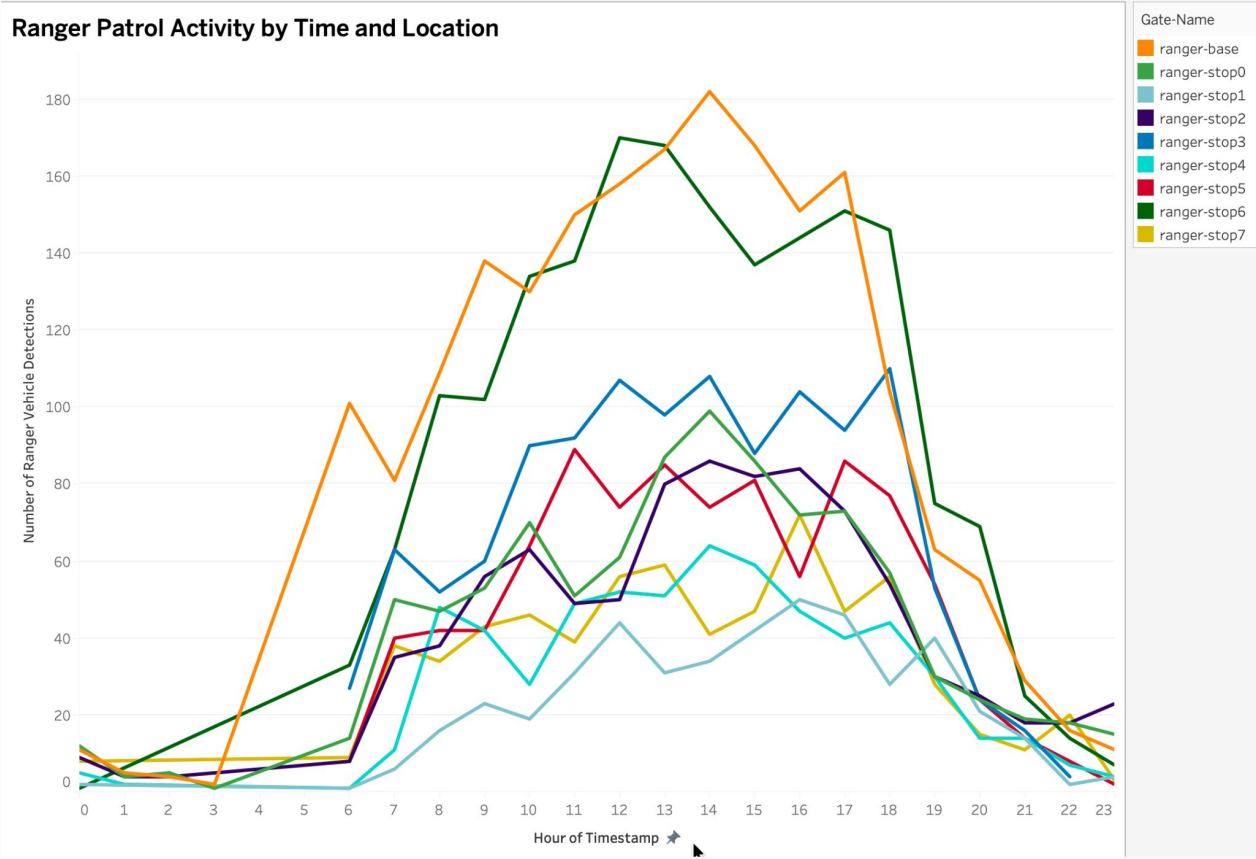
Hypothesis:

- Key points to consider:
  - The same vehicle ID was used throughout the entire period which means this was probably a long term ID given to the vehicle.
  - The vehicle did not enter a camping gate but stayed for 4-5 days consistently.
  - There was a very strict and consistent pattern to the time of stay, start time, and end time.

- Hypothesis:
  - Due to the pattern and unique behavior of the vehicles, this may be related to conservation studies being conducted in the preserve or some kind of work being done.

Pattern of Life: Ranger Patrol and Maintenance Operations

- **Vehicles:** Park service vehicles (2P)
- **Spatial Activity:** Ranger vehicles repeatedly visit multiple ranger-stops and the ranger base, which correspond to restricted road endpoints and operational hubs within the preserve. Activity is distributed across many ranger-managed locations rather than concentrated at a single site.
- **Temporal Activity:** Ranger activity follows a strong daily rhythm, beginning in the early morning, peaking during late morning and afternoon hours, and tapering off in the evening. Overnight activity is minimal. Individual locations show repeated but brief detections rather than extended stays.
- **Hypothesis:** This pattern represents routine ranger patrols, inspections, and maintenance work conducted during normal operational hours. The coordinated timing across multiple ranger-stops and the ranger base is consistent with structured daily work schedules rather than visitor movement.



Potential Suspicious Activity

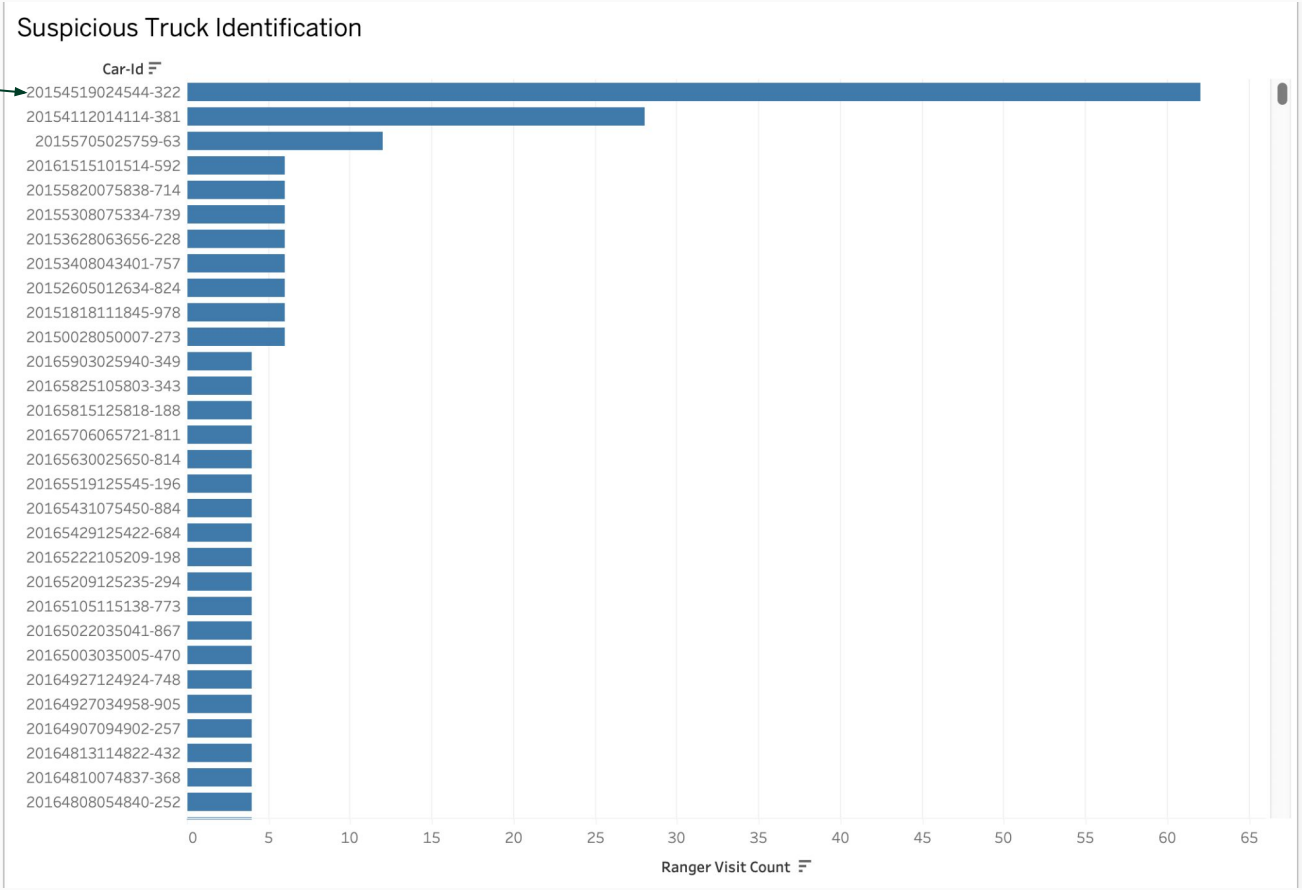
- Only authorized rangers should be allowed to enter through “Gates”
- Most ranger vehicles don’t fall under any of the car types (Cause they have P appended to the end of their cartype)
- Issue: There are **46** instances of a **car type 4** entering through gates 3, 5, 6, and ranger-stops 3 and 6
- Explanation for high number of entries of ranger-stop 0 and 2:
  - Those stops are located between 2 general gates and are not restricted by “Gates”.
  - There is an issue with ranger stops 3 and 6 because these should be restricted by “Gates”.

Car Types @ Ranger Stops and Authorized-Only Gates

	Car-Type						
Gate-Name	1	2	2P	3	4	5	6
gate0			124				
gate1			124				
gate2			440				
gate3			1,188		46		
gate4			916				
gate5			1,625		46		
gate6			1,139		46		
gate7			670				
gate8			1,996				
ranger-base			1,996				
ranger-stop0	6,327	3,815	948	2,515	731	501	241
ranger-stop1	12		454				
ranger-stop2	6,327	3,815	889	2,515	731	501	241
ranger-stop3			1,190		46		
ranger-stop4			612				
ranger-stop5			920				
ranger-stop6			1,832		46		
ranger-stop7			672				

Identification of Suspicious Heavy Truck Activity at Ranger-Stop Locations

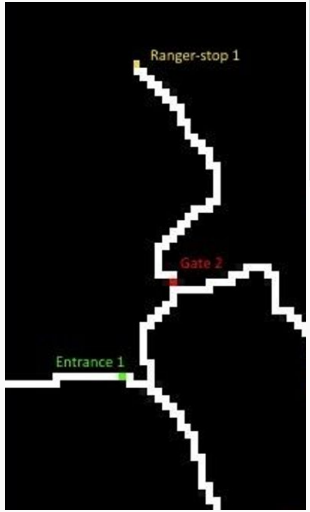
This Vehicle ID was identified as the most suspicious vehicle due to repeated multi-day visits to ranger-stop sensors, heavy truck classification, and absence of ranger authorization, consistent with illicit dumping activity described in the investigation.

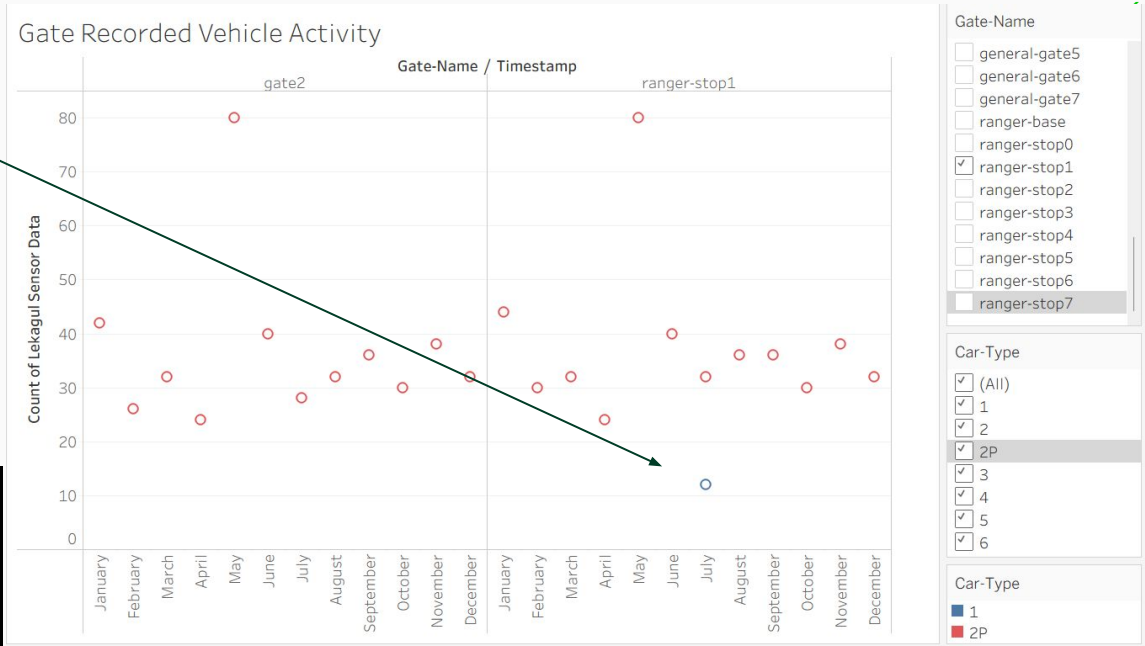


Unusual Outlier in Ranger-Stop1

This specific car type (1) was recorded at ranger-stop1, which can only be reached after passing through gate2. However, there are no records of any car type 1 passing through gate2 at any instance.

This opens up the possibility that there are other ways to enter through specific areas without being detected by the sensors. Possibly through another route outside the path, or bypassing gate2 sensors. This car type was detected in stop1 12 times in July alone.





Next Steps

- Continue digging to find out which location the illegal dumping may be occuring.
  - Try to identify the route the truck is taking.
- Is there a specific time that the dumping is occurring?
  - Night/ Day
  - Days of week
- Try to find other patterns in the data.

Questions